

Kohat University of Science & Technology

Institute of Computing

Kohat, Khyber Pakhtunkhwa, Pakistan



BSCS / BSSE Final Year Project (FYP)

Intelligent Pharmacy Design Document

Submitted By:

Sr.No	Registration No	Name	Class	Email
1	CS120222110	Talha Rehman	BSCS 7th B, Open	talhakhank51@gmail.com
2	CS120222063	Basit	BSCS 7th B, Open	basit.ahmed.usman@gmail.com

1. Introduction

1.1 Purpose

The IPSMS is designed to modernize traditional pharmacy management by integrating Artificial Intelligence. It serves two distinct user groups: Pharmacy Owners (Vendors) and Patients (Users). The system aims to reduce financial losses caused by expired medicine while acting as a personal healthcare assistant for patients through AI-driven prescription reading and health monitoring.

1.2 Scope

- **Inventory Management:** Tracking stock levels, batch expiries, and predicting demand.
- **AI Healthcare Assistant:** Using Computer Vision and NLP to interpret prescriptions, suggest alternatives, and set dosage reminders.
- **Platform:** A cross-platform Mobile Application (Flutter) and Web Dashboard (React.js).

2. System Architecture

The system follows a **Three-Tier Architecture** leveraging Supabase for backend services and Google's Gemini API for intelligence.

2.1 Architectural Diagram

2.2 Components

1. **Client Presentation Layer:**
    - **Mobile App (Flutter):** Primary interface for Patients and Store Owners.
    - **Web Dashboard (React.js + Tailwind CSS):** Admin and heavy-duty inventory management for vendors.
2. **Middle Layer (API & Logic):**
    - **Supabase Client:** Handles direct secure communication between the client and the database.
    - **AI Processing:** Requests sent to **Gemini API** for Optical Character Recognition (OCR) and Natural Language Processing (NLP) on prescriptions.
3. **Data & Services Layer:**
    - **Authentication:** Supabase Auth (User, Vendor, Admin roles).
    - **Database:** PostgreSQL (hosted via Supabase).
    - **Storage:** Cloud storage for prescription images.

3. Module Design

3.1 Module 1: Vendor (Store Owner)

This module focuses on business logic and inventory control.

- **Inventory Management:** CRUD operations for medicines. The system tracks medicine_batches to manage expiry dates specifically.

- **Stock Alerts:** The alerts table triggers notifications for LOW_STOCK or EXPIRY events based on predefined thresholds.

- **Sales & Orders:** Management of incoming digital orders via the orders table.

3.2 Module 2: User (Patient) with AI

This module focuses on health accessibility and adherence.

- **Smart Prescription Reader:**
  - **Input:** User uploads an image (stored in prescriptions.image_url).
  - **Process:** Gemini API analyzes the image.
  - **Output:** ai_extracted_text and ai_summary are stored and displayed to the user in simple language.

- **Medicine Reminders:** Users can set schedules (e.g., "8:00 AM daily") stored in the reminders table.

- **Alternative Suggestions:** If a prescribed medicine is out of stock, the system suggests generic alternatives using the medicines catalog.

4. Data Design (Database Schema)

The database is implemented in **PostgreSQL**. Below is the entity relationship analysis based on the provided SQL schema.

4.1 Key Entities

Entity	Description	Key Attributes
Users	Central identity for all actors.	id, email, role ('user', 'vendor', 'admin', 'doctor'), password_hash
Pharmacies	Linked to a Vendor (user_id). Stores store metadata.	id, vendor_id, license_number, city
Medicines	The master catalog of drugs (Global).	id, name, generic_name, strength
Batches	Specific stock instances linked to a Pharmacy.	batch_number, expiry_date, quantity, selling_price
Prescriptions	Stores user uploads and AI results.	image_url, ai_extracted_text, ai_summary
Orders	Transactional data between User and Pharmacy.	status ('pending', 'paid', 'completed'), total_amount

4.2 Critical Relationships

- **Inventory Tracking:** Pharmacies 1:N Medicine_Batches. This allows multiple pharmacies to sell the same Medicine but with different expiry dates and stock levels.
- **Audit Trail:** Medicine_Batches 1:N Stock_Transactions. Every time stock is added (IN) or sold (OUT), a record is created in stock_transactions to prevent theft/loss.

- **Security:** Users 1:N OTPs and Password_Reset_Tokens handles secure access and verification.

5. Technology Stack

- **Frontend (Mobile):** Flutter (Dart)
- **Frontend (Web):** React.js, Tailwind CSS
- **Backend Services:** Supabase (PostgreSQL, Auth, Realtime) (Note: Proposal mentions Node.js/Express, which may be used for custom server-side logic or AI middleware).
- **Artificial Intelligence:**
  - **LLM/Vision:** Google Gemini API.
- **Tools:** GitHub (Version Control), Postman (API Testing).